

Name _____ Teacher _____



GOSFORD HIGH SCHOOL

2013

HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK 2

MATHEMATICS – EXTENSION 1

Duration- 90 minutes plus 5 minutes reading time

Section 1 Multiple choice	5 questions worth 1 mark each. (Answer this section on the multiple choice response sheet provided)	/5
Section 2 Question 6	Angle between 2 lines, Iterative methods, Integration	/15
Question 7	Locus, Integration	/15
Question 8	Integration, Mathematical Induction, Graphing	/15
TOTAL		/50

SECTION 1: MULTIPLE CHOICE

Questions 1 – 5 are to be answered on the multiple choice answer sheet provided. Each question is worth 1 mark.

Question 1. The exact value of $\int_0^{\frac{\pi}{4}} \sin^2 x dx$ is:

- A) $\frac{\pi}{8} + \frac{1}{4}$ B) $\frac{\pi}{4} - \frac{1}{2}$
C) $\frac{\pi}{8} - \frac{1}{4}$ D) $\frac{\pi}{4} + \frac{1}{2}$

Question 2. Using the substitution $u = 1 - x^3$, $\int \frac{x^2}{\sqrt{1-x^3}} dx$ is equivalent to:

- A) $-3 \int \frac{du}{\sqrt{u}}$ B) $-\frac{1}{3} \int \frac{du}{\sqrt{u}}$
C) $-\frac{1}{3} \int \frac{du}{u^2}$ D) $3 \int \frac{du}{u^2}$

Question 3. The curve $y = \frac{(2x+3)(x-2)}{(x-3)(x+1)}$, has:

- A) Vertical asymptotes at $x = 3$ and $x = -1$ and a horizontal asymptote at $y = 2$.
B) Vertical asymptotes at $x = -3$ and $x = 1$ and a horizontal asymptote at $y = 2$.
C) Vertical asymptotes at $x = 3$ and $x = -1$ and horizontal asymptotes at $y = -\frac{3}{2}$ and $y = 2$.
D) Vertical asymptotes at $x = 3$ and $x = -1$ and a horizontal asymptote at $y = 0$.

Question 4. The parabola with parametric equations $x = 2t + 1$ and $y = t^2 - 2$, has:

- A) Vertex $(-1, 2)$ and focus $(-1, 3)$
- B) Vertex $(-1, 2)$ and focus $(0, 3)$
- C) Vertex $(1, -2)$ and focus $(1, -3)$
- D) Vertex $(1, -2)$ and focus $(1, -1)$

Question 5. When using mathematical induction to prove

$$1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{2}, \text{ step 3 would read:}$$

Prove true for $n = k + 1$, if true for $n = k$, ie prove:

- A) $\frac{3^k - 1}{2} + 3^{k+1} = \frac{3^{k+1} - 1}{2}$
- B) $\frac{3^k - 1}{2} + 3^k = \frac{3^{k+1} - 1}{2}$
- C) $\frac{3^k}{2} + 3^{k+1} = \frac{3^{k+1} - 1}{2}$
- D) $\frac{3^k - 2 + 3^{k+1}}{2} = \frac{3^{k+1} - 1}{2}$

SECTION 2: FREE RESPONSE

Questions 6, 7 and 8 are to be answered on your own paper. Start each question on a new page. All necessary working must be shown. Marks may not be awarded for untidy or poorly set out work.

Question 6.

- a) Find the acute angle between the lines $y = 3x - 2$ and $y = 2 - x$, giving your answer to the nearest degree. (2)
- b) Find the acute angle between the curves $y = 3x^3 - 2$ and $y = (x - 2)^2$ at their point of intersection $(1, 1)$. Give answer correct to the nearest degree. (3)
- c) Let $f(x) = x^3 + 5x^2 + 17x - 10$. The equation $f(x) = 0$ has only one real root. Given that the root lies between 0 and 2:
 - i) Use one application of the 'halving the interval' method to find a smaller interval containing the root. (2)
 - ii) Which end of the interval found in part i) is closer to the root? Justify your answer. (2)
- d) Use Newton's method to find a second approximation to the positive root of $x - 2 \sin x = 0$. Take $x = 1.7$ as the first approximation. Give answer correct to 2 decimal places. (3)
- e) Evaluate $\int_0^1 6x\sqrt{9 - x^2} dx$, using the substitution $u = 9 - x^2$. Leave your answer in surd form. (3)

Question 7. Start a new page.

- a) Find the equation of the chord of contact of the tangents to the parabola $x^2 = y$ from the point $(-1, -5)$. (3)
- b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The equation of the tangents at P and Q respectively are $y = px - ap^2$ and $y = qx - aq^2$.
 - i) The tangents at P and Q meet at the point T. Show that the coordinates of T are $(a(p + q), apq)$. (2)
 - ii) Find the equation of the locus of T if PQ is a focal chord. (2)

SOLUTIONS

- c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. The chord PQ subtends a right angle at the origin. M is the mid point of PQ.

i) Show that $pq = -4$ (2)

ii) Show that the locus of M is given by $x^2 = 2a(y - 4a)$ (3)

d) Find $\int x\sqrt{x+1} dx$, using the substitution $u^2 = x+1$. (3)

Question 8. Start a new page.

- a) The region under the curve $y = \cos x + \sec x$, above the x axis and between $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x axis. Show that the volume of the solid formed is $\frac{5\pi(\pi+2)}{8}$ cubic units. (4)

- b) Prove by mathematical induction that $3^n \geq 1 + 2n$ for all integers n , $n \geq 1$. (4)

- c) Consider the function $f(x) = \frac{x^4 + 3x^2}{x^4 + 3}$.
- i) Show that $f(x)$ is an even function. (1)
- ii) What is the equation of the horizontal asymptote to the graph $y = f(x)$? (1)
- iii) Find the x coordinates of all stationary points for the graph $y = f(x)$? (3)
- iv) Sketch the graph $y = f(x)$. You are not required to find any points of inflexion. (2)

SECTION 1: Multiple choice

$$\begin{aligned} 1) \int_0^{\pi/4} \sin^2 x \, dx &= \frac{1}{2} \int_0^{\pi/4} 1 - \cos 2x \, dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4} \\ &= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right] \\ &= \frac{\pi}{8} - \frac{1}{4} \end{aligned} \quad \text{--- (C)}$$

$$\begin{aligned} 2) \int \frac{x^2}{\sqrt{1-x^3}} \, dx &= -\frac{1}{3} \int \frac{-3x^2}{\sqrt{1-x^3}} \, dx \quad u = 1-x^3 \\ &= -\frac{1}{3} \int \frac{du}{\sqrt{u}} \quad \frac{du}{dx} = -3x^2 \end{aligned} \quad \text{--- (B)}$$

$$\begin{aligned} 3) y &= \frac{(2x+3)(x-2)}{(x-3)(x+1)} \\ \lim_{x \rightarrow \infty} \frac{2x^2 - 2x - 6}{x^2 - 2x - 3} &= \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2} - \frac{2x}{x^2} - \frac{6}{x^2}}{\frac{x^2}{x^2} - \frac{2x}{x^2} - \frac{3}{x^2}} \\ &= 2 \end{aligned} \quad \begin{aligned} (x-3)(x+1) &\neq 0 \\ \therefore x &\neq 3, -1 \\ \text{i.e. vertical asymptotes} & \quad x=3 \text{ and } x=-1 \end{aligned}$$

i.e. horizontal asymptote $y=2$

--- (A)

4) $x = 2t + 1$, $y = t^2 - 2$

$$2t = x - 1$$

$$t = \frac{x-1}{2}$$

$$y = \left(\frac{x-1}{2}\right)^2 - 2$$

$$4y = (x-1)^2 - 8$$

$$(x-1)^2 = 4y + 8$$

$$(x-1)^2 = 4(y+2)$$

$$\therefore \text{vertex } (1, -2)$$

$$\text{focus } (1, -1)$$

— (D)

5) $1 + 3 + 3^2 + \dots + 3^{k-1} + 3^k = \frac{3^{k+1} - 1}{2}$

$$\frac{3^k - 1}{2} + 3^k = \frac{3^{k+1} - 1}{2}$$

— (B)

SECTION 2

6) a) $y = 3x - 2$

$$M_1 = 3$$

$$y = 2 - x$$

$$M_2 = -1$$

$$\tan \theta = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right|$$

$$= \left| \frac{3 + 1}{1 - 3} \right|$$

$$= \left| -\frac{3}{2} \right|$$

$$\tan \theta = \frac{3}{2}$$

$$\theta = 56^\circ$$

b) $y = 3x^3 - 2$

$$y' = 9x^2$$

$$\text{When } x = 1,$$

$$M_1 = 9$$

$$y = (x-2)^2$$

$$y' = 2(x-2)$$

$$\text{When } x = 1,$$

$$M_2 = -2$$

$$\tan \theta = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right|$$

$$= \left| \frac{9 + 2}{1 - 18} \right|$$

$$= \frac{11}{17}$$

$$\theta = 33^\circ$$

c) $f(x) = x^3 + 5x^2 + 17x - 10$

i) $f(0) = -10 < 0$ $f(2) = 8 + 20 + 34 - 10 = 52 > 0$

$$f(1) = 1 + 5 + 17 - 10$$

$$= 13 > 0$$

$\therefore$ root lies between 0 and 1 (i.e. $0 < x < 1$)

should include 'since there is a sign change from 0 to 1, and it is continuous.'

ii) $f\left(\frac{1}{2}\right) = \frac{1}{8} + \frac{5}{4} + \frac{17}{2} - 10$

$$= -\frac{1}{8}$$

$\therefore$ root lies between $\frac{1}{2}$ and 1

$\therefore$ 1 is closer to the root than zero.

$$d) f(x) = x - 2 \sin x \quad f(1.7) = 1.7 - 2 \sin 1.7 \\ = -0.2833$$

$$f'(x) = 1 - 2 \cos x$$

$$f'(1.7) = 1 - 2 \cos 1.7 \\ = 1.2577$$

$$a_1 = \frac{a - f(a)}{f'(a)} \\ = 1.7 + \frac{0.2833}{1.2577} \\ = 1.93$$

$$e) \int_0^1 6x \sqrt{9-x^2} dx$$

$$\text{let } u = 9 - x^2$$

$$\text{when } x=0, u=9$$

$$\frac{du}{dx} = -2x$$

$$x=1, u=8$$

$$du = -2x dx$$

$$\begin{aligned} -3 \int_0^1 -2x \sqrt{9-x^2} dx &= -3 \int_9^8 u^{1/2} du \\ &= -3 \left[\frac{2u^{3/2}}{3} \right]_9^8 \\ &= -2 \left[8^{3/2} - 9^{3/2} \right] \\ &= -2 \left[16\sqrt{2} - 27 \right] \end{aligned}$$

$$7. a) x^2 = y \quad (-1, -5)$$

$$xx_1 = 2a(y+y_1)$$

$$x \cdot -1 = 2 \cdot \frac{1}{4} (y-5)$$

$$-x = \frac{1}{2} (y-5)$$

$$-2x = y-5$$

$$y = -2x + 5$$

$$b) P(2ap, ap^2), Q(2aq, aq^2) \quad x^2 = 4ay$$

$$i) px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p-q) = a(p-q)(p+q)$$

$$x = a(p+q)$$

$$\text{sub into } y = px - ap^2$$

$$y = ap(p+q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$$\therefore T \text{ has coordinates } (a(p+q), apq)$$

$$ii) \text{ If } PQ \text{ is a focal chord, } pq = -1$$

$$\therefore y = apq$$

$$y = -a \quad \text{ie locus is the directrix}$$

c) $P(2ap, ap^2)$ $Q(2aq, aq^2)$ $x^2 = 4ay$

i) If PQ subtends a right angle at the origin, then

$$M_{OP} \cdot M_{OQ} = -1$$

$$M_{OP} = \frac{ap^2}{2ap} = \frac{p}{2}$$

$$M_{OQ} = \frac{aq^2}{2aq} = \frac{q}{2}$$

$$\therefore \frac{p}{2} \cdot \frac{q}{2} = -1$$

$$pq = -4$$

ii) Midpoint $x = \frac{2ap+2aq}{2} = \frac{a(p+q)}{2}$

$$y = \frac{aq^2+ap^2}{2} = \frac{a(q^2+p^2)}{2}$$

$$x^2 = a^2(p+q)^2$$

$$= a^2(p^2+q^2+2pq)$$

$$= a^2(p^2+q^2-8)$$

$$\frac{x^2}{a^2} = p^2+q^2-8$$

$$p^2+q^2 = \frac{x^2}{a^2} + 8$$

now $y = \frac{a(q^2+p^2)}{2}$

$$= \frac{a\left(\frac{x^2}{a^2} + 8\right)}{2}$$

$$2y = \frac{x^2}{a} + 8a$$

$$x^2 = 2ay - 8a^2$$

$$x^2 = 2a(y-4a)$$

d) $\int x\sqrt{x+1} dx$

let $u^2 = x+1$

$$x = u^2 - 1$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$\int x\sqrt{x+1} dx = \int (u^2-1)u \cdot 2u du$$

$$= \int 2u^4 - 2u^2 du$$

$$= 2\left[\frac{u^5}{5} - \frac{u^3}{3}\right]$$

$$= 2\left[\frac{(\sqrt{x+1})^5}{5} - \frac{(\sqrt{x+1})^3}{3}\right] + c$$

8) a) $y = \cos x + \sec x$

$$V = \pi \int_0^{\pi/4} (\cos x + \sec x)^2 dx$$

$$= \pi \int_0^{\pi/4} \cos^2 x + 2\cos x \sec x + \sec^2 x dx$$

$$= \pi \int_0^{\pi/4} \cos^2 x + 2 + \sec^2 x dx$$

$$= \pi \int_0^{\pi/4} \frac{1}{2}(1+\cos 2x) + \sec^2 x + 2 dx$$

$$= \pi \left[\frac{1}{2}x + \frac{1}{4} \sin 2x + \tan x + 2x \right]_0^{\pi/4}$$

$$= \pi \left[\frac{\pi}{8} + \frac{1}{4} \sin \frac{\pi}{2} + \tan \frac{\pi}{4} + \frac{\pi}{2} - 0 \right]$$

$$= \pi \left[\frac{\pi}{8} + \frac{1}{4} + 1 + \frac{\pi}{2} \right]$$

$$= \pi \left[\frac{5\pi + 10}{8} \right]$$

$$= \frac{5\pi(\pi+2)}{8} \text{ cubic units}$$

b) Prove $3^n \geq 1+2n$, $n \geq 1$

Step 1: Show true for $n=1$

$$\begin{array}{ll} \text{LHS} = 3^1 & \text{RHS} = 1+2 \\ = 3 & = 3 \end{array}$$

$$\text{LHS} \geq \text{RHS}$$

$\therefore$ true for $n=1$

Step 2: Assume true for $n=k$, k an integer

$$\text{i.e. } 3^k \geq 1+2k$$

Step 3: Prove true for $n=k+1$, if true for $n=k$

$$\text{i.e. Prove } 3^{k+1} \geq 1+2(k+1)$$

$$3^{k+1} \geq 3+2k$$

$$\text{now } 3^{k+1} = 3 \cdot 3^k$$

$$\text{and } 3 \cdot 3^k \geq 3(1+2k) \text{ from assumption}$$

$$3^{k+1} \geq 3+6k$$

now for $k \geq 1$, $6k > 2k$

$$\therefore 3^{k+1} \geq 3+2k$$

$\therefore$ true for $n=k+1$, if true for $n=k$

$\therefore$ proven true by mathematical induction

c) $f(x) = \frac{x^4 + 3x^2}{x^4 + 3}$

i) $f(-x) = \frac{(-x)^4 + 3(-x)^2}{(-x)^4 + 3}$

$$= \frac{x^4 + 3x^2}{x^4 + 3}$$

$$= f(x) \quad \therefore \text{even fn.}$$

ii) $\lim_{x \rightarrow \infty} \frac{\frac{x^4}{x^4} + \frac{3x^2}{x^4}}{\frac{x^4}{x^4} + \frac{3}{x^4}}$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{3}{x^2}}{1 + \frac{3}{x^4}}$$

$$= 1$$

$\therefore$ horizontal asymptote is $y=1$